LEVERS

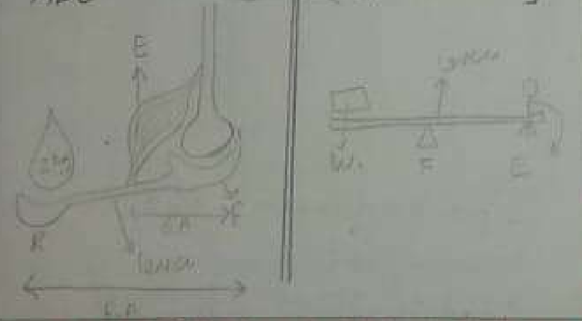
Descriptions :-

→ Lever is a rigid body which rotates around the fixed axis
→ axis of Rotation is called Fulcrum
→ two forces acting along the length of the lever at particular distance are Efforts and weight / load / Resistance.
→ Efforts tends to produce movement while Resistance Resist movement.
→ Effort Arm is is the perpendi distance from the fulcrum to weigh
→ Resistance Arm is the perpendi distance from the fulcrum to the Effort Arm

Terms Related to Levers :-

Lever → Bony Segment
fulcrum → Joint
Effort → muscle contraction.
weight → Resistance against Effort.
Effort Arm (EA) → Distance from fulcrum to Effort.
Resistance Arm → Distance from fulcrum to Resistance.

ADL ⟷ Human body



CLASSIFICATION OF LEVERS

→ On the basis of position of fulcrum, Efforts & Resistance

Remember as free
F F E

1st Order lever | 2nd order lever | 3rd Order lever

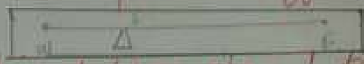
1st Order lever

also known as lever of stability.

→ In this types, fulcrum lies between centre & Resistance. It may either situated in centre or toward Efforts or Resistance

Conditions

• when fulcrum is then resistance arm is equal to Effort Arm.



• When fulcrum placed towards the Resistance arm the Resistance Arm is lesser the Effort Arm.

• when fulcrum is placed towards Effort the Effort arm is lesser then Resistance Arm.

MECHANICAL ADVANTAGE

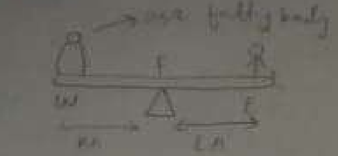
$MA = \frac{EA}{RA}$ [Ratio of Effort arm & Resistance arm]

$MA > 1 \longrightarrow$ Advantage
$MA < 1 \longrightarrow$ Disadvantage

Mechanical Advantage for 1st Order

• mechanical Advantage in 1st Order lever can be 1, greater or lesser than 1.
• if $EA = RA$ then it is Equilibrium
• if $EA > RA$ then Advantage
• if $RA > EA$ then Disadvantage.

ADL

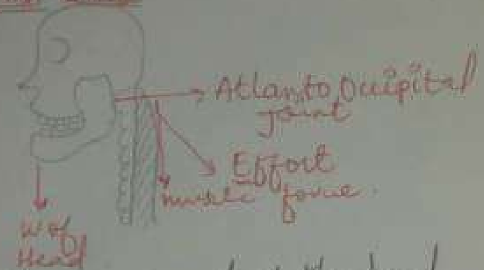


See saw

Scissor.

Cutting plier

HUMAN BODY



- Nodding movement of the head
- Lever = Skull
- fulcrum = Atlantooccipital joint
- weight = COG of head
- Efforts = Insertion point of posterior neck muscle

2nd Order lever

→ In this type resistance lies between Efforts & Resistance
→ Effort arm is always greater than Resistance arm
→ also called as lever of power or lever of Advantage

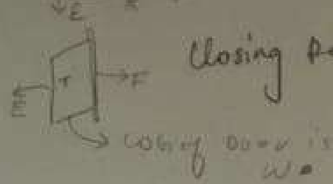
Mechanical Advantage

EA is always greater in 2nd order lever ∴ In 2nd Order lever mechanical Advantage takes place

ADL

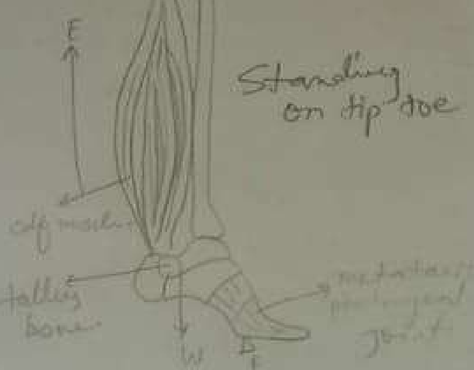
Shoulder wheel

nut cutter.



Closing Door

Human body



Standing on tip toe

- plantarflexion of Ankle
- lever = foot
- Ankle Joint = fulcrum
- weight: COG of foot
- Effort = E: Insertion point of plantar flexor ie gastrocnemius and Soleus.

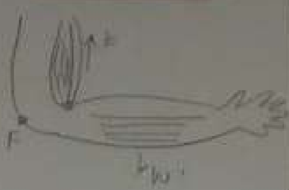
3rd Order lever

- In this type Effort lies between Weight and fulcrum
- mechanical Advantage
- This lever is known as lever of Disadvantage
- RA is greater than EA
- RA > EA

ADL

fisherman tool

HUMAN BODY



[It plays imp role in Exercise also, if patient is weak then we prescrible short lever Exercise like patient is suffering from a shoulder pain & weak also & could not able to Abduct his shoulder As it is long lever the we prescribe short lever Exercise like ~~abfirst~~ flex Elbow the to shoulder Abduction. In this case Resistance arm decreases

Flexion of Elbow by biceps brachii →

lever = forearm and hand
fulcrum = Elbow joint
Weight = COG of forearm & hand
insertion point of Biceps brachii at radial tuberosity

Postures and Direction of Movement

Supine

Someone in the supine position is lying on his or her back.

Supine

Prone

Someone in the prone position is lying face down.

Prone

Right Lateral Recumbent

The Right lateral recumbent, or RLR, means that the patient is lying on their right side.

Right Lateral Recumbent

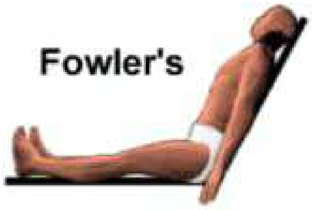
Left Lateral Recumbent

The left lateral recumbent, or LLR, means that the patient is lying on their left side.

Left Lateral Recumbent

Fowler's Position

A person in the Fowler's position is sitting straight up or leaning slightly back. Their legs may either be straight or bent.



Fowler's

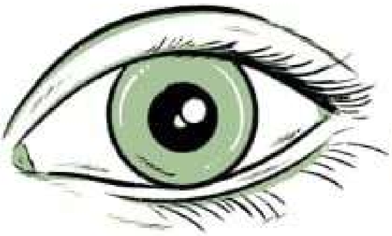
Trendelenberg Position

A person in the Trendelenberg position is lying supine with their head slightly lower than their feet.

Trendelenberg

Subdivisions of living anatomy

Inspection

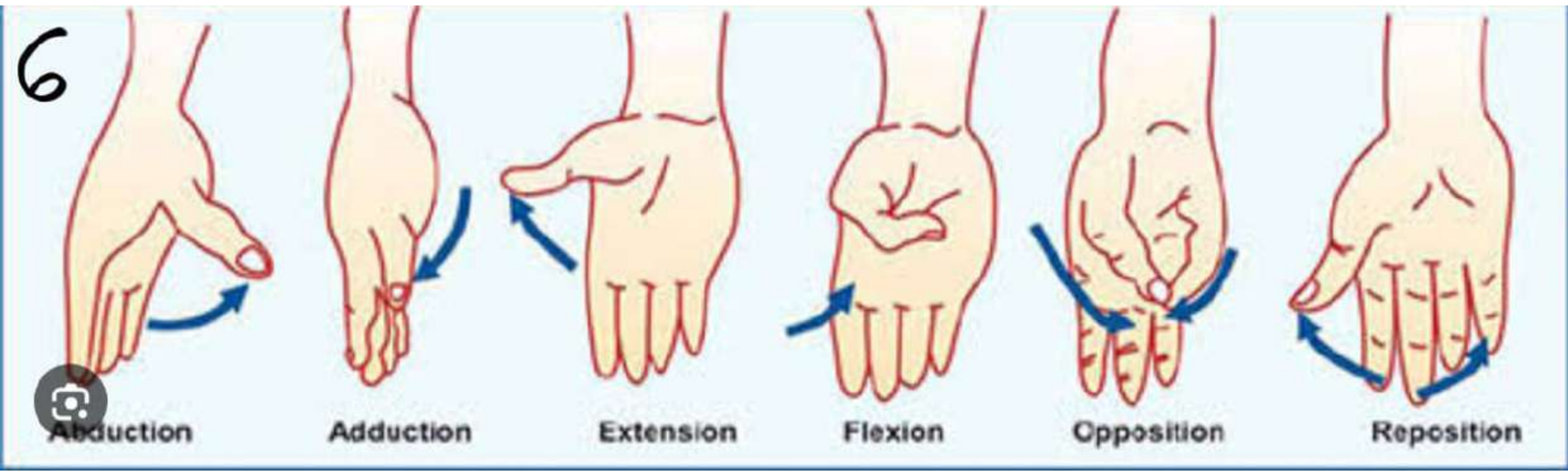


Palpation

Percussion

Auscultation

6



Abduction
Adduction
Extension
Flexion
Opposition
Reposition